

Dhruv Tomar

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Education

University of Illinois Urbana-Champaign

Bachelor of Science in Integrative Biology

Bachelor of Science in Computer Science

Expected Graduation: May 2027

GPA: 3.86/4.00

Relevant Coursework: Theoretical Biology & Models, Probability & Statistics, Cells Tissues & Development, Ecology, Physiology, Genetics, Evolution, Behavioral Ecology, Ornithology, Deep Learning for Computer Vision, Linear Algebra, Algorithms

Research Experience

University of Illinois Urbana-Champaign, Department of Natural Resources & Environmental Sciences

Undergraduate Research Assistant

August 2024 - Present

Advisor: Dr. Benjamin Van Doren

- Investigating the effects of light pollution on bird flight behavior using 3-D thermal video tracking in the Van Doren Lab of Migration Biology.
- Optimized background subtraction methods for thermal video object tracking.
- Trained and optimized a YOLOv8 machine learning model to improve object detection performance.
- Performing statistical analyses for over 10,000 tracks using MATLAB and Python (NumPy, pandas, SciPy, TensorFlow), studying various aspects of flight kinematics such as directionality, turning, and acceleration.
- Experience setting up acoustic monitoring stations in the field.

Cornell University, Boyce Thompson Institute

Research Intern

June 2026 - August 2026

Advisor: Dr. Edward Buckler

- Summer 2026 REU Intern at the Boyce Thompson Institute.
- Benchmarked a fine-tuned DNA language model (GeneCAD) against reference plant genome annotations using bioinformatics tools such as BUSCO, AGAT, OrthoFinder, and ESMFold structural confidence.
- Built a CAFE5 gene-family evolution pipeline spanning ~20 angiosperm species to infer lineage-specific expansions and contractions.
- Studied the evolution and transcription of young genes in 26 *Zea mays* genomes using ATAC-seq and RNA-seq data.

University of California San Diego, Department of Computer Science

Research Intern

June 2025 - August 2025

Advisors: Dr. Ryan Kastner & Dr. Curt Schurgers

- Summer 2025 REU Intern in the Engineers for Exploration program.
- Collaborated with National Geographic explorers on two large-scale bioacoustic data mining projects, characterizing soundscapes with little to no labeled data.

- Applied template matching and unsupervised deep learning to build a Graph Neural Network pipeline uncovering latent links between audio features and environmental variables.
- Pioneered a modular Python package enabling scalable, cross-dataset bioacoustic analysis and integration into future ecological research workflows using libraries such as PyTorch, TensorFlow, and Librosa.

Publications

Peer-Reviewed Publications

Perry, S., Zhang, T., Kamboj, S., Jajodia, A., **Tomar, D.**, & Kastner, R. (2026). Studying domain shifts in bioacoustic recording style between focal and soundscape recordings with acoustic indices. *Ecological Informatics*, 95, 103739. DOI: <https://doi.org/10.1016/j.ecoinf.2026.103739>

Science Writing

Tomar, D. (2026). Recent advances in phylogenetics through the advent of modern computation. *Double Helix Digest*, 2(1), 17-24.

Tomar, D. (2025). Factors affecting migratory bird-window collisions and effective mitigation strategies. *Double Helix Digest*, 1(1), 9-16.

Conference & Symposium Presentations

Reef Futures, Riviera Maya, Mexico, December 2026. Kamboj, S., Perry, S., Schiffman, P., Jajodia, A., **Tomar, D.**, Zhu, V., Rodríguez-Troncoso, P., Alvarado, J., Schurgers, C., Kastner, R. "Evaluating Bioacoustics as Next-Generation Reef Monitoring: Feasibility of Moving Beyond Traditional Surveys to Evaluate Reef Health" (non-speaking contributor, accepted).

American Ornithological Society, Amherst, MA, August 2026. **Tomar, D.**, Maron, M., Corcoran, A., Van Doren, B. "Three-dimensional thermal video tracking reveals light-induced behavioral disruptions in nocturnal migrants" (poster).

- **Won Nellie Johnson Baroody Award for best presentation**

George and Helen Kohut Scientific Symposium, Ithaca, NY, August 2026. **Tomar, D.**, Liu, Z. "A DNA foundation model enables artifact-controlled analysis of young genes across 26 *Zea mays* genomes" (poster).

Undergraduate Research Symposium, Urbana, IL, April 2026. **Tomar, D.**, Corcoran, A., Van Doren, B. "Three-dimensional thermal video tracking reveals light-induced behavioral disruptions in nocturnal migrants" (oral presentation).

Undergraduate Research Symposium, Urbana, IL, April 2025. **Tomar, D.**, Corcoran, A., Van Doren, B. "3D Tracking in Thermal Videos to Analyze the Effects of Light Pollution on Birds" (poster).

Honors and Awards

Nellie Johnson Baroody Award for Best Presentation, American Ornithological Society	2026
Engineering Visionary Scholarship (\$4,000)	2026
American Ornithological Society Travel Grant (\$600)	2026
Chester W. and Nadine C. Houston Scholarship (\$1,000)	2026
Illinois for Illinois (I4I) Study Abroad Scholarship (\$1,000)	2025
James Scholar Honors Program	2025

Leadership & Service Experience

Double Helix Digest

Writing Chair

June 2026 - Present

- Write review articles for the *Double Helix Digest*, an undergraduate biology journal.
- Assist writers with reading and synthesizing research papers.

University of Illinois Urbana-Champaign, Grainger College of Engineering

Lead Engineering Learning Assistant

January 2026 - Present

- Teaching and mentoring a section of ~20 Computer Science students in a first-year engineering orientation (ENG 100) on academic, career, and campus life topics.
- Facilitating student learning and community-building using curated lesson materials.
- Holding weekly instructional meetings to review pedagogy and curriculum design.

Illini Wildlife and Conservation Club

President / Vice President / Treasurer

May 2024 - Present

- Organize monthly general meetings and invite professors to discuss their research.
- Manage funds and budget for a 50+ member club.
- Participate in philanthropy such as turning off lights around campus to conserve energy and picking up litter around campus.
- Volunteer with migratory bird banding efforts by the Illinois Natural History Survey.

UIUC Bird Strike Survey

Bird Collision Monitor

September 2023 - Present

- Survey 20+ buildings on campus during the bird migration season several times a week for bird-window collisions following survey protocols and log them via ArcGIS QuickCapture.
- Collect specimens for use in teaching and research.

Technical Skills

Programming: C++, Python, R, MATLAB, C, Java, JavaScript, Verilog

ML & Scientific Python: NumPy, pandas, scikit-learn, PyTorch, TensorFlow, Matplotlib, Librosa

Tools & Systems: Git, SSH, Vim, Bash, PowerShell, Docker, PostgreSQL, Bootstrap 5, MongoDB, HPC, SLURM

Professional Memberships

American Ornithological Society (2023-Present)

Association for Computing Machinery (2025-Present)